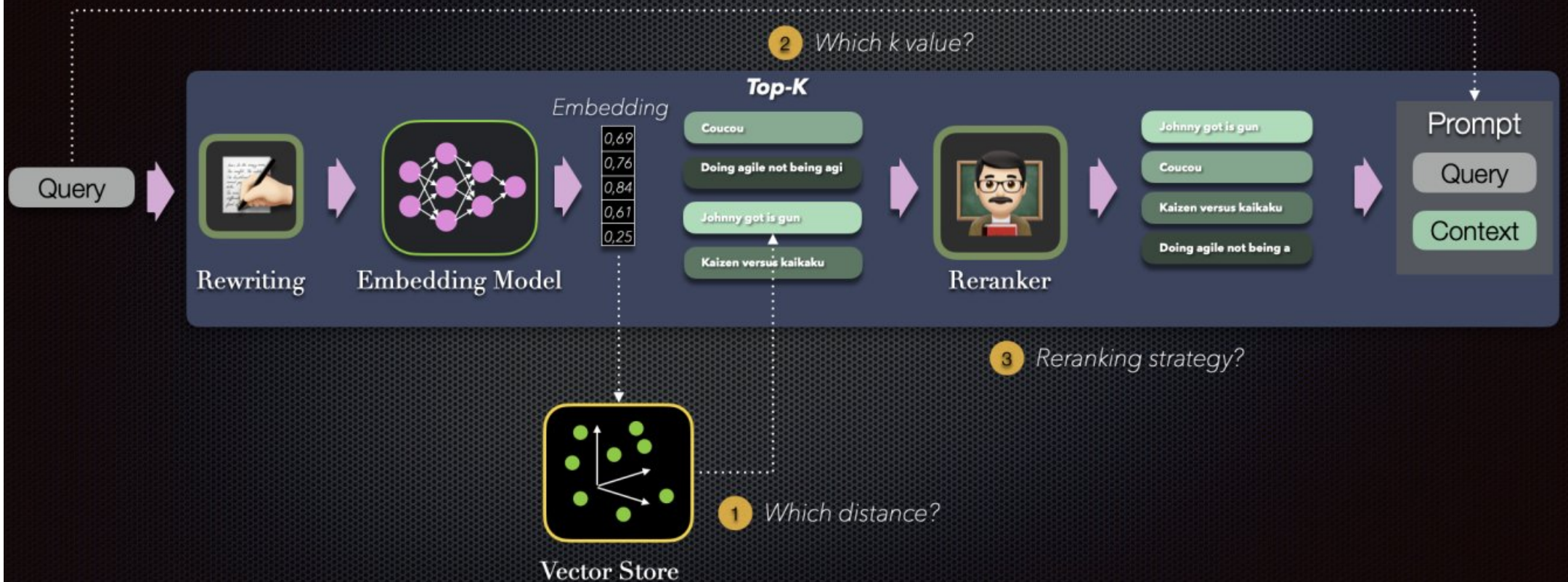
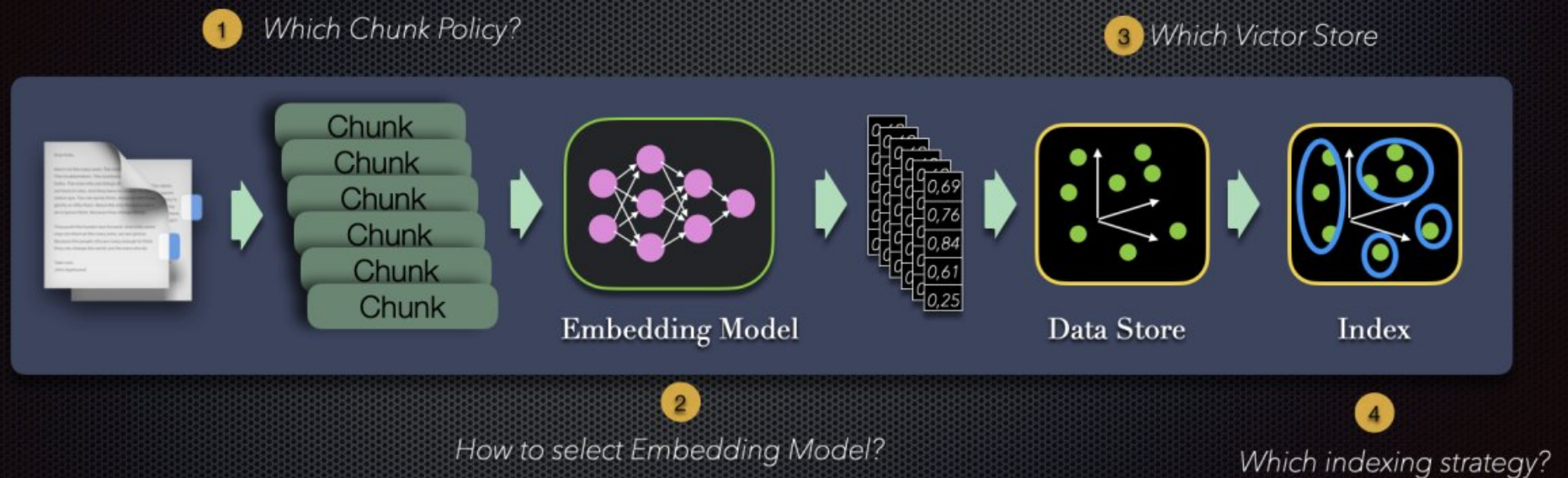


Generation Pipeline Challenges

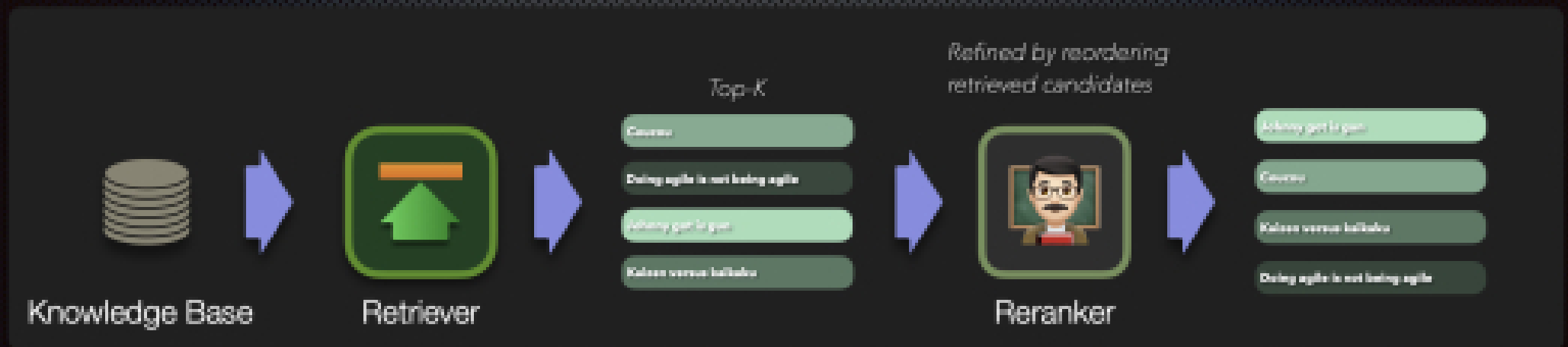


Ingestion Pipeline Challenges



Reranking

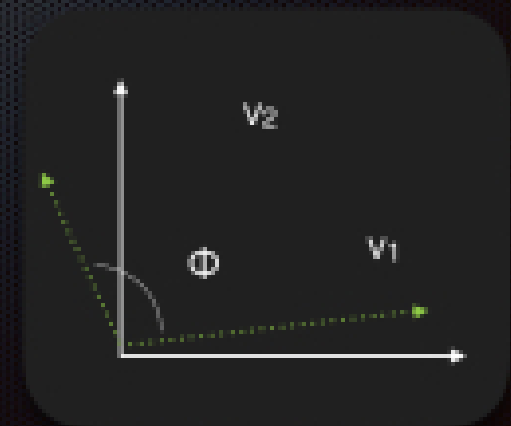
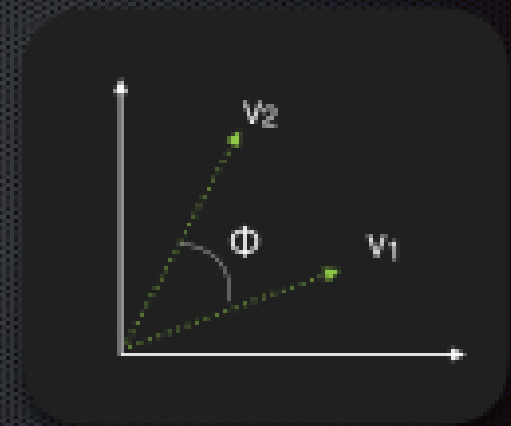
- Reranking is a refinement step that reorders retrieved candidates using a more computationally expensive but more accurate model,
 - often a cross-encoder or LLM-based scorer.



Embedding Models

act as the semantic bridge between natural language and efficient numerical retrieval mechanisms.

- An embedding model is a **neural model trained** to convert text (or other modalities) into dense numerical vectors that encode semantic meaning. Its objective is to
 - map **semantically similar** inputs to **nearby points** in vector space
 - while pushing unrelated concepts farther apart.



```
from langchain.text_splitter import CharacterTextSplitter

text = "Why did the scarecrow win an award? Because he was outstanding in his field! This is the example text for this fun exercise."
text_splitter = CharacterTextSplitter(chunk_size=35, chunk_overlap=2, separator=' ')
documents = text_splitter.create_documents([text])

# Print the resulting chunks
for doc in documents:
    print(doc.page_content)
```

Embedding

- Embedding is the transformation of textual (or multimodal) content into dense numerical vectors that capture semantic meaning in a high-dimensional space.

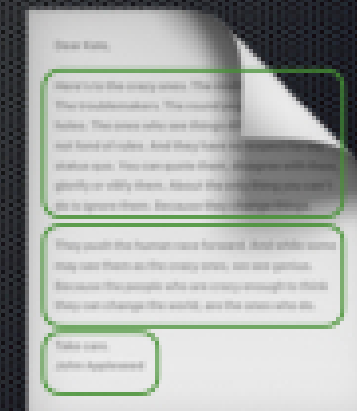


Chunking

- Chunking is the process of splitting source documents into smaller, semantically coherent units called chunks
 - before embedding and indexing.
- The goal is to balance context granularity and retrieval efficiency.
- Chunks must be
 - **large enough** to preserve meaning,
 - **small enough** to avoid dilution during similarity search

strategies may be

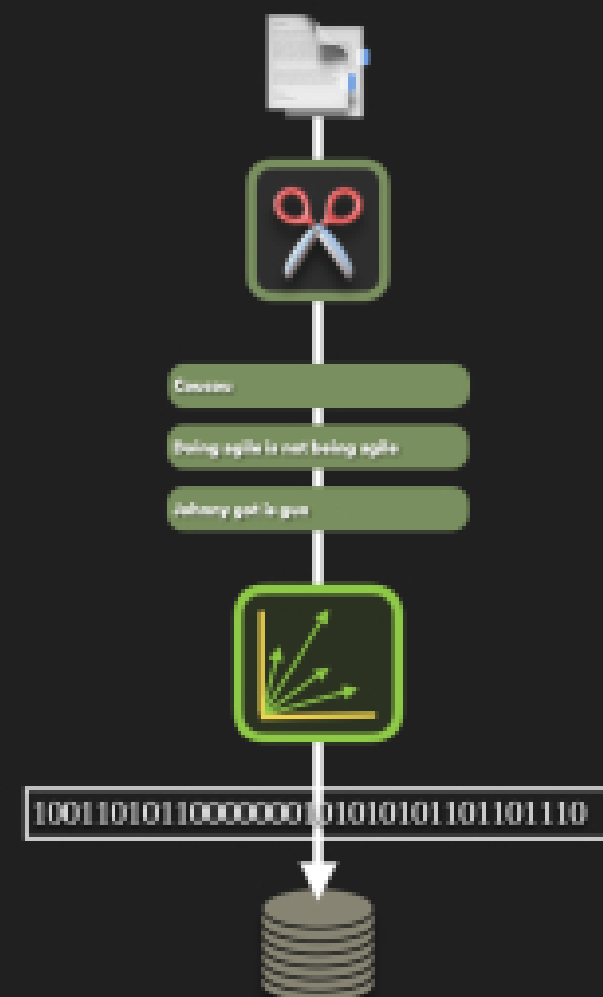
- **fixed-size**,
- **structure-aware** (headings, sections),
- **semantic** (topic boundaries), or
- **sliding-window** based



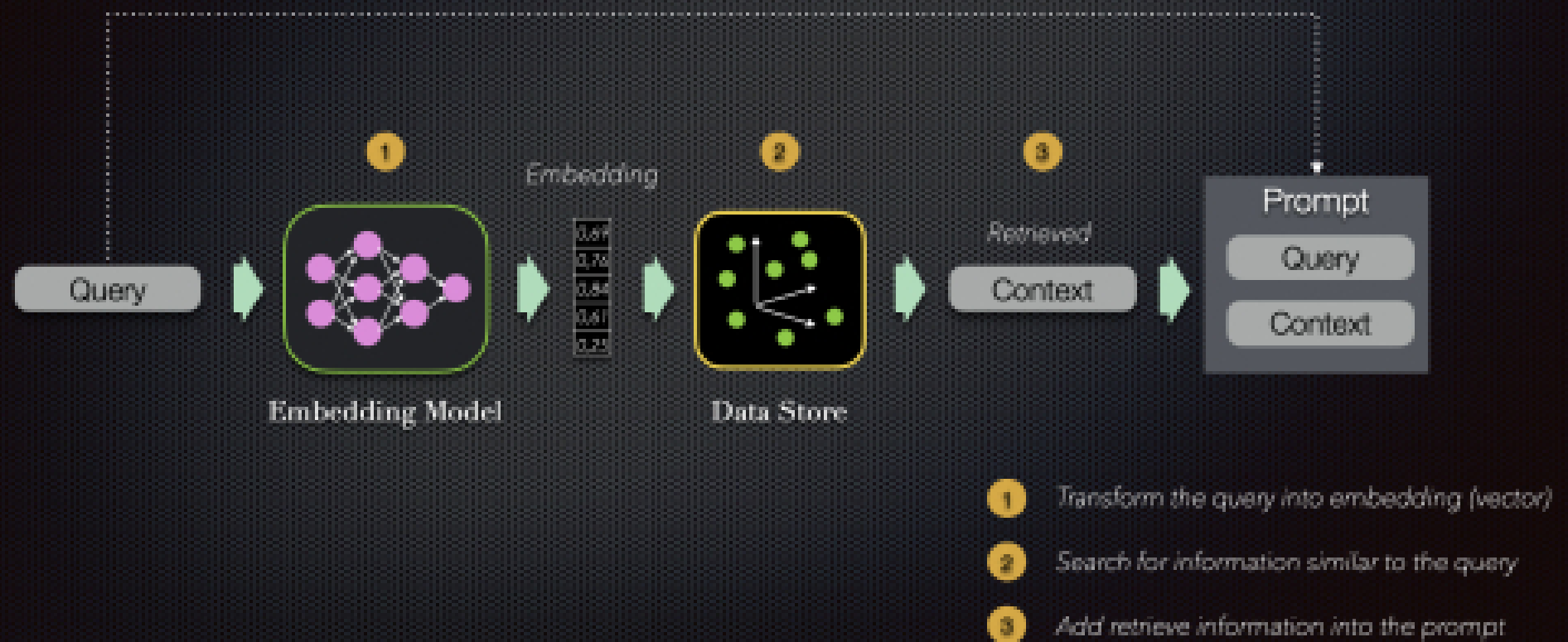
they directly impact recall, precision, latency, and cost in a RAG system.

Ingestion Pipeline

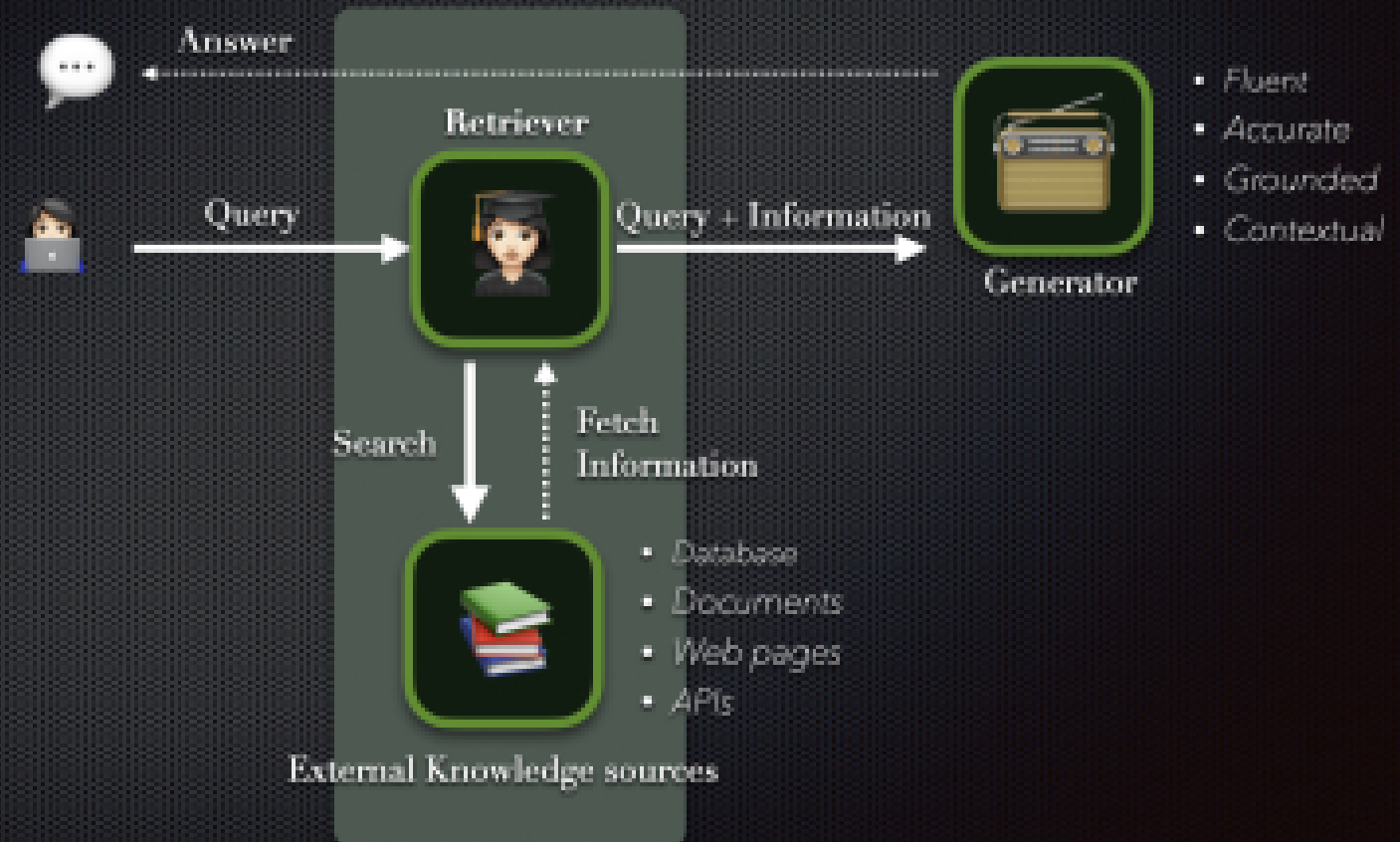
- The ingestion pipeline is the offline or asynchronous process that prepares data for retrieval. It typically includes:
 - document acquisition,
 - cleaning and normalization,
 - structural or semantic chunking,
 - metadata enrichment,
 - embedding generation, and
 - indexing into a vector or hybrid store.



Generation Pipeline



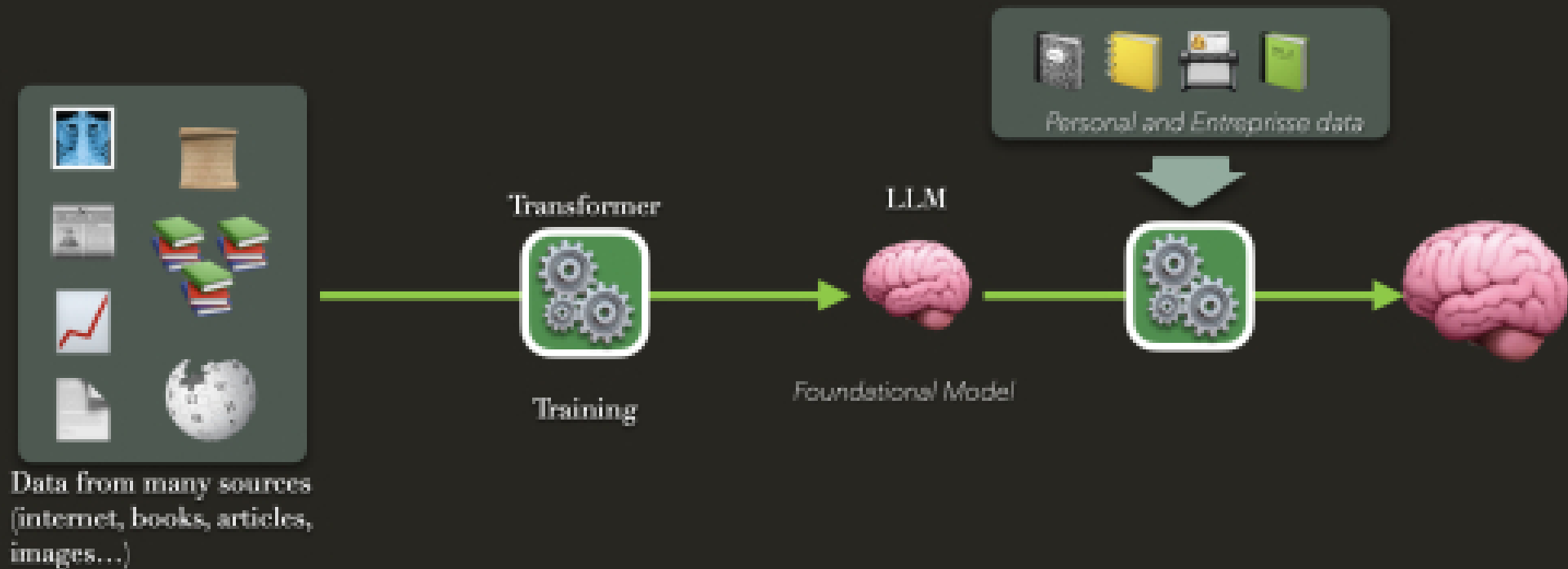
RAG



Advantage

- up-to-date information
- Consistency across queries
- More nuanced answer
- Credibility by referencing sources
- Adaptability to new informations
- Transparency (clear sources)
- Allows for fact-checking

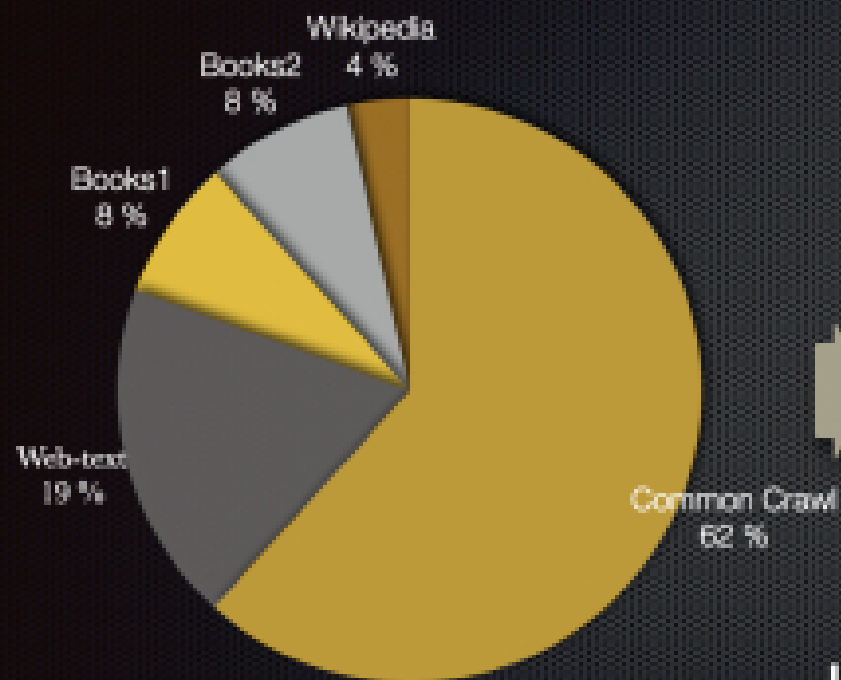
GPT - Fine Tuning



Foundation Model's issues

- They are missing **private** and **undisclosed** data such as
 - Enterprise financial
 - Individual and personal data
- They are **limited** to data they have been trained on
 - The cut-off
- They don't **update** with real time information

With a little help from my LLM



The Model is even much more bigger than what you could expect from your own trained model



LLM are approximatively trained on a equivalent of 450.000 years human reading

With a little help from my LLM

The Model is even much more bigger
than what you could expect from your
own trained model



*Use the Model to infer
(classification, regression...)*

Before LLM

